Heater PCB Design Iterations

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Why PCB Design for a Heater?

A printed circuit board is designed to simplify and combine multiple components within a system to have a robust and resistant design. Specifically for my project I'm controlling an infrared heater I have designed an all-in-one PCB that has the circuitry to control the heater with your own local Wi-Fi network.

Three Control Components

Within figure 1 it shows the three main components being the computer (Raspberry Pi Zero 2W), power supply and the high load relays. These are the three elements that are to be condensed down into a single PCB. Also, it will allow for an overall smaller footprint.

One way the PCB can be condensed is by simplifying the high load relays since it has more features than what's needed by having a NO (normally open) and NC (normally closed) terminal where I only need a NO terminal.

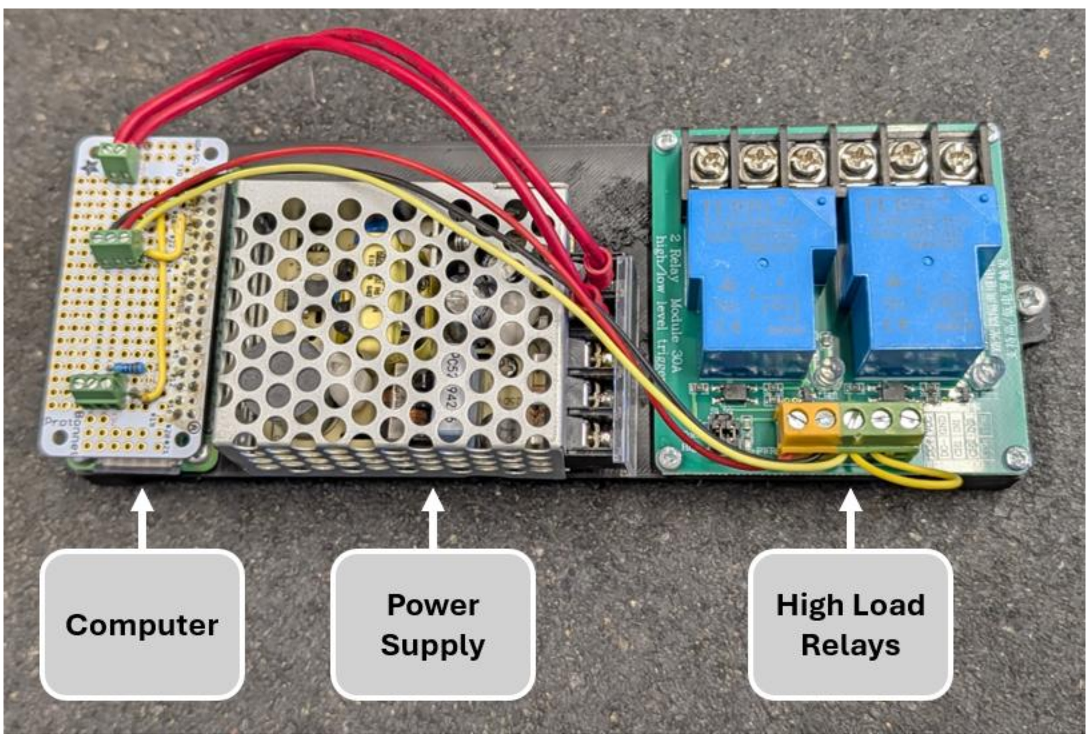

Figure 1: Three Main Components

Relay PCB

The relay PCB was designed to start learning about PCB design software and to downsize the overall footprint of the relays to make the overall design smaller.

Version 1.0

Dimensions (Length x Width): 76mm x 57.5mm

Number of Layers: Two

Software: EasyEDA Online

Description:

The first relay PCB that I designed was in EasyEDA online since it was a simple PCB software that conveniently had many components, footprints and models from JLCPCB already integrated. This made ordering the PCB with components very simple since the gerber files and the JLCPCB order would all be done in one button press from the EasyEDA online software.

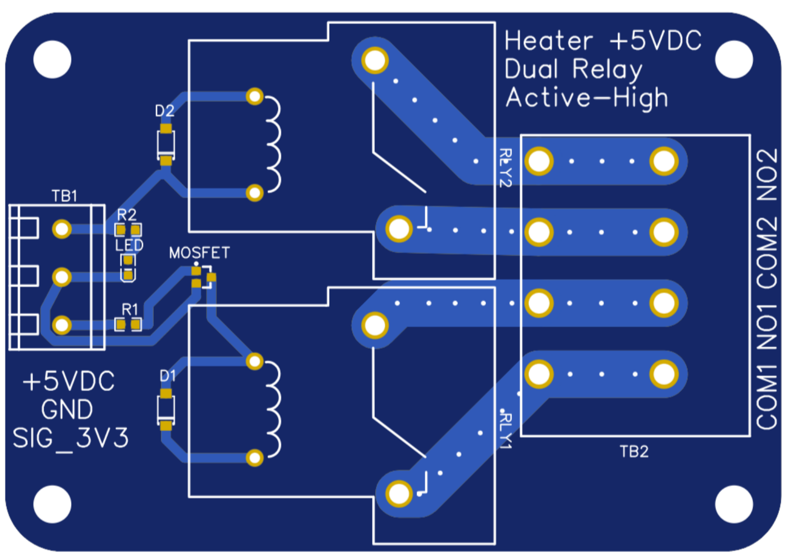

Figure 2: Relay PCB v1.0

Version 1.1

Dimensions (Length x Width): 76mm x 57.5mm

Number of Layers: Two

Software: Kicad 9.0

Description:

This 1.1 version relay PCB is very similar to version one except now I started using an open-source software named Kicad. This allows me to make much more complex designs seen below with nice mounting holes that are properly sized for appropriate M3 screws. Kicads learning curve was a little steep but there were many useful tutorials online. My favorite part of this new design is the QR code leading to the public GitHub repository that is the project associated with this boards use.

Figure 3: Relay PCB v1.1 All Board Components

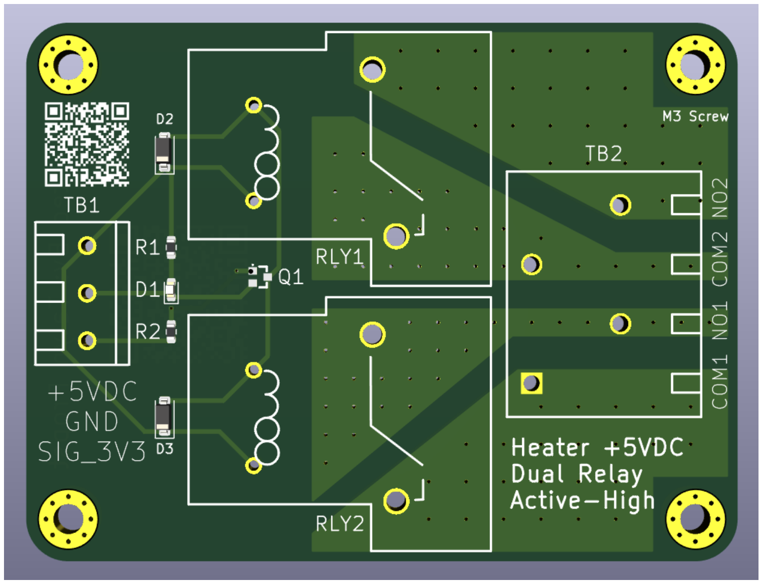

Figure 4: Relay PCB v1.1 Front

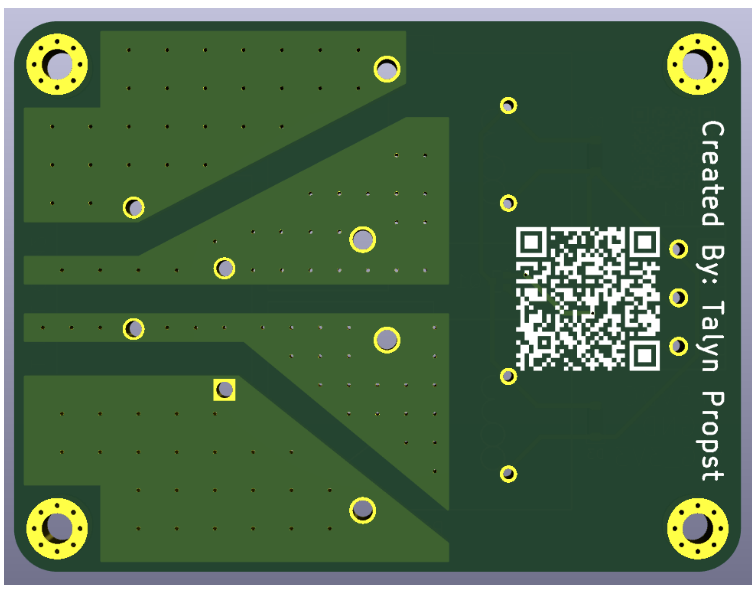

Figure 5: Relay PCB v1.1 Back

Version 1.2

Dimensions (Length x Width): 76mm x 57.5mm

Number of Layers: Two

Software: Kicad 9.0

Description:

Version 1.2 is identical to version 1.1 except for learning a valuable lesson when I tested version 1.1 about adding a pull-down gate resistor. A pull-down gate resistor was needed because when I connected the board to power with +5VDC and GND (0VDC) then the 3.3v signal was not connected and the N-type MOSFET would be in a floating state flickering the relays partially on and off very fast sounding like a taser. Though when the 3.3v signal was connected this stopped as the MOSFET gate now had a known reference voltage.

While not a major issue the board can operate but it was addressed because it was unsatisfactory and unprofessional when hooking up the board with power and ground first making the relays sound like a taser also, this could have damaged a load if connected to the output of the relays. Within figure 6 you can see there is an additional resistor (R3) for the MOSFET gate pull down.

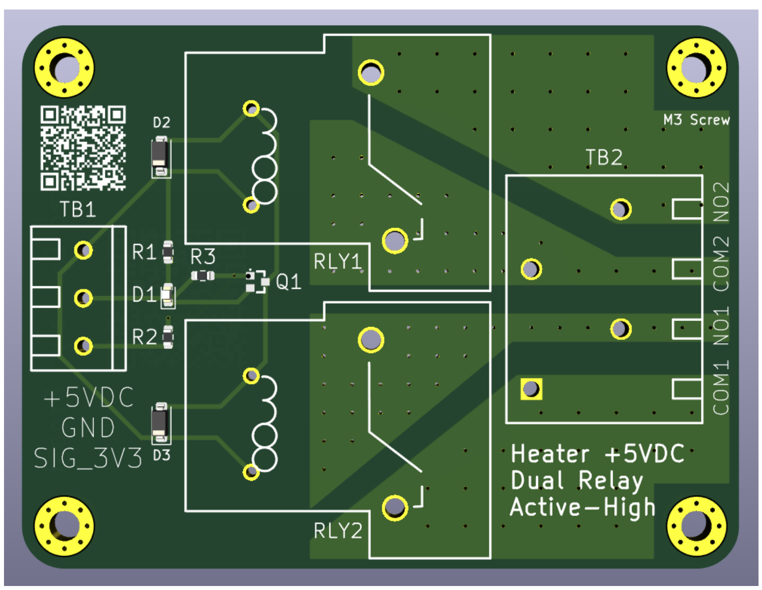

Figure 6: Relay PCB v1.2 Front

Full System PCB

A full system PCB design was started shortly after version 1.1 of the PCB relay after it was tested and worked showing my calculations were correct. The goal of this all-in-one PCB is to have a smaller footprint, less hookup wires and robust circuitry.

Version 1.0

Dimensions (Length x Width): 100mm x 88mm

Number of Layers: Four

Software: Kicad 9.0

Description:

Version 1.0 of the full system PCB has a major focus on high voltage AC isolation from low voltage DC to prevent interference. Originally, I wanted to design a power supply but upon reading a book the author recommended if someone sells what you need just buy their product and avoid the complex calculations of switch mode power supplies. Glad I did because the Meanwell MPM-20-5 has worked great for converting +240VAC to +5VDC to power the relays and computer on the full system PCB. The relays on the board are the same design as from the relay PCB one change is the removal of solder mask on the bottom layer seen in figure 9. This is to enhance cooling of the heavy amperage load the relays and terminal block pass between each other. One other major change to this board is that its four layers and 1oz copper weight allowing for efficient energy transfer and heat dissipation of the high amperage load.

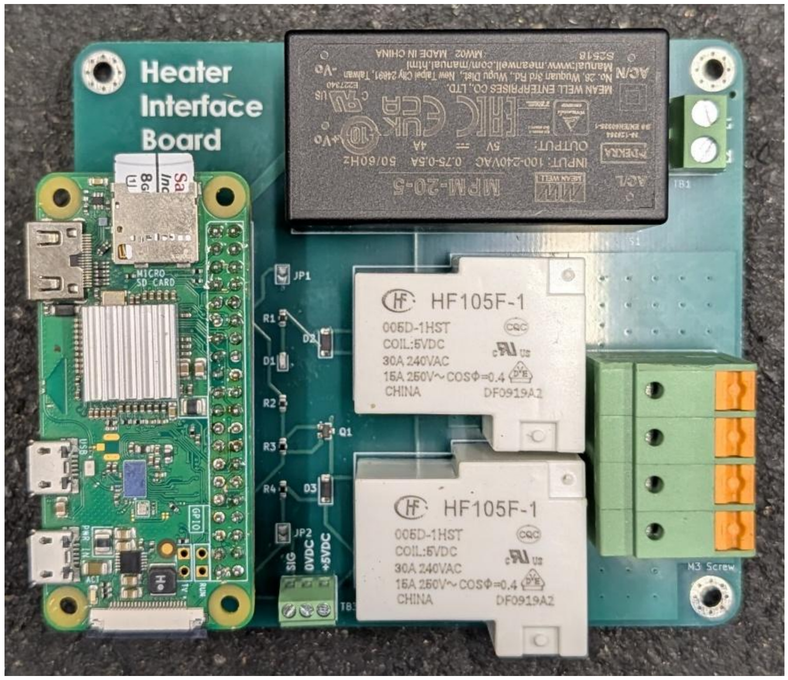

Figure 7: Full System PCB All Board Components

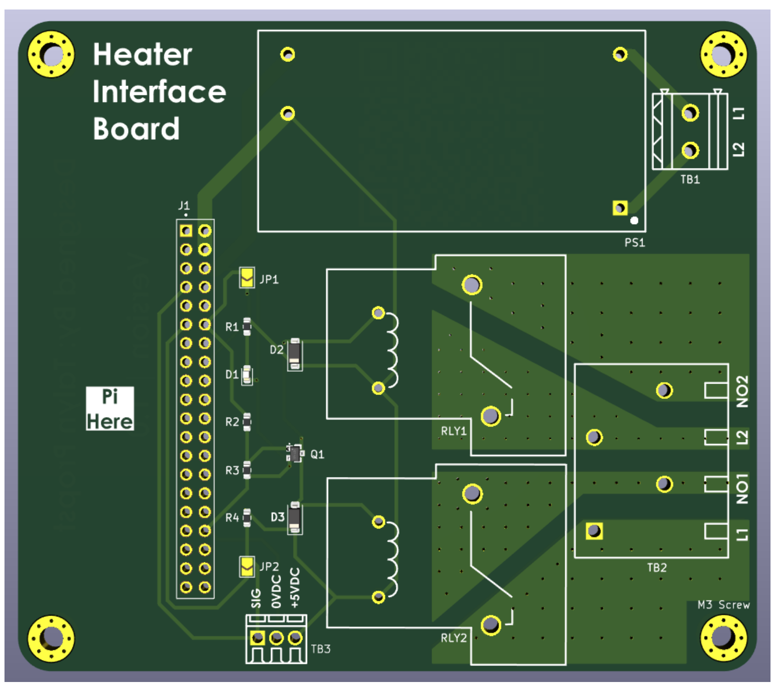

Figure 8: Full System PCB v1.0 Front

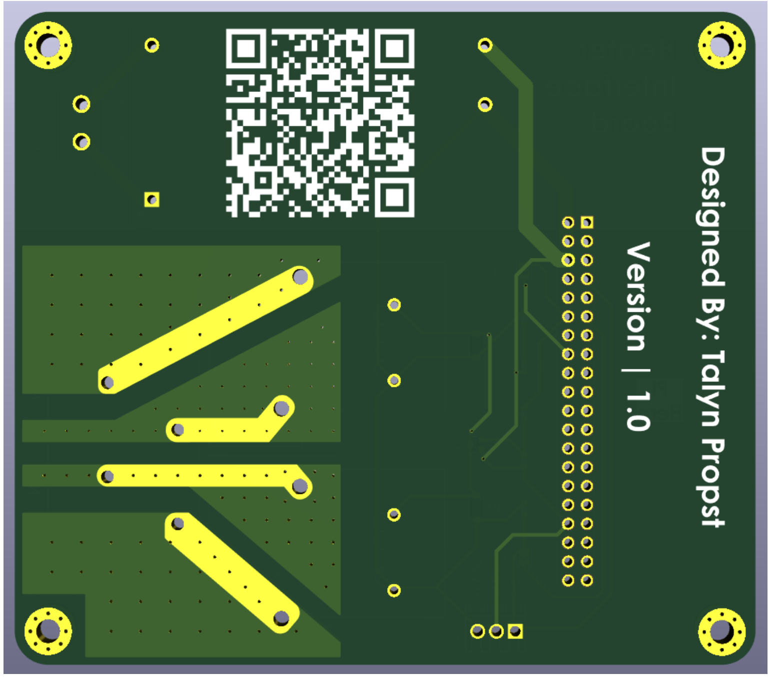

Figure 9: Full System PCB v1.0 Back

Version 1.1

Dimensions (Length x Width): 100mm x 88mm

Number of Layers: Four

Software: Kicad 9.0

Description:

Version 1.1 comes with new trace routing and two more jumper pads. All four jumper pads are used for connecting and disconnecting a general-purpose input output (GPIO) pin to be able to use a backup GPIO pin in case of a failure. The problem fixed of the trace routing had nothing to do with the actual schematic is happened that for the backup relay pin that its signal also ran through the original pin and if that happened to fail and it was permanently stuck in a low or high state the backup pin would have not been able to change the relay state due to its routing.

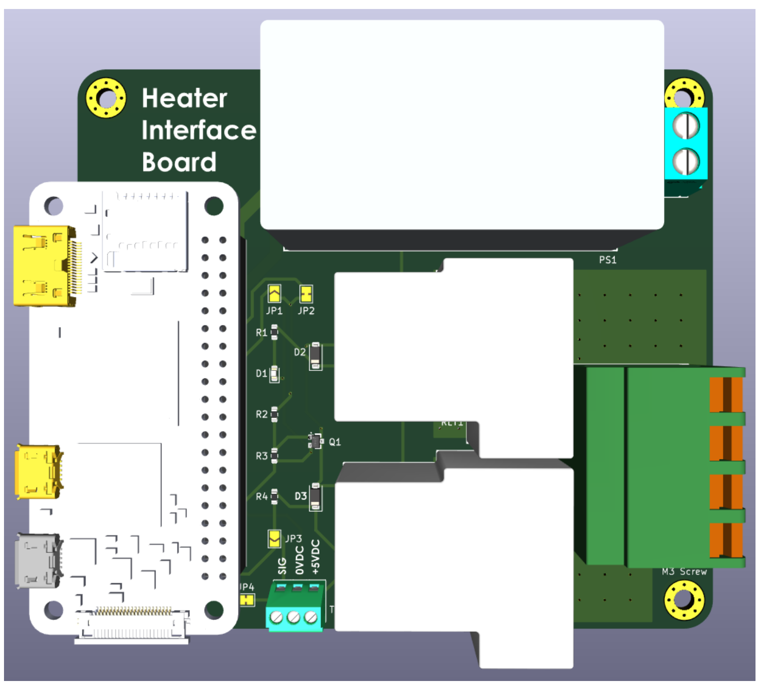

Figure 10: Full System PCB v1.1 Front